Name:Manjyot Wadhwa
Roll no :65
Div:D15C

Experiment 1

Implementation of Linear and Logistic Regression

1. Dataset Source

Dataset Name: **Heart Disease Dataset**
Source: Kaggle
Link: https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

The dataset was downloaded from Kaggle and used in CSV format.

---

2. Dataset Description

The dataset contains medical attributes used to determine the presence of heart disease.

📊 Dataset Size

- 303 samples
- 14 attributes

📌 Features Include:

- age
- sex
- cp (chest pain type)
- trestbps (resting blood pressure)
- chol (cholesterol)
- fbs (fasting blood sugar)
- restecg (resting ECG results)
- thalach (maximum heart rate)
- exang (exercise induced angina)
- oldpeak
- slope
- ca (number of major vessels)
- thal
- target (heart disease presence)

🎯 Targets Used in This Experiment

To apply both regression and classification using the same dataset:

- **Linear Regression** → Predict `chol` (cholesterol level – continuous)
- **Logistic Regression** → Predict `target` (0 = no disease, 1 = disease)

Thus, a single Kaggle dataset was used for both models.

3.Mathematical Formulation

🔹 Linear Regression

Linear Regression models:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$$

Objective:

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

Where:

- (y) = actual cholesterol value
- (\hat{y}) = predicted value

🔹 Logistic Regression

Logistic Regression uses the sigmoid function:

$$P(y = 1) = \frac{1}{1 + e^{-z}}$$

Where:

$$z = \beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n$$

Loss Function: Binary Cross Entropy

---

4. Algorithm Limitations

Linear Regression

- Assumes linear relationship
- Sensitive to outliers
- Cannot model non-linear patterns

Logistic Regression

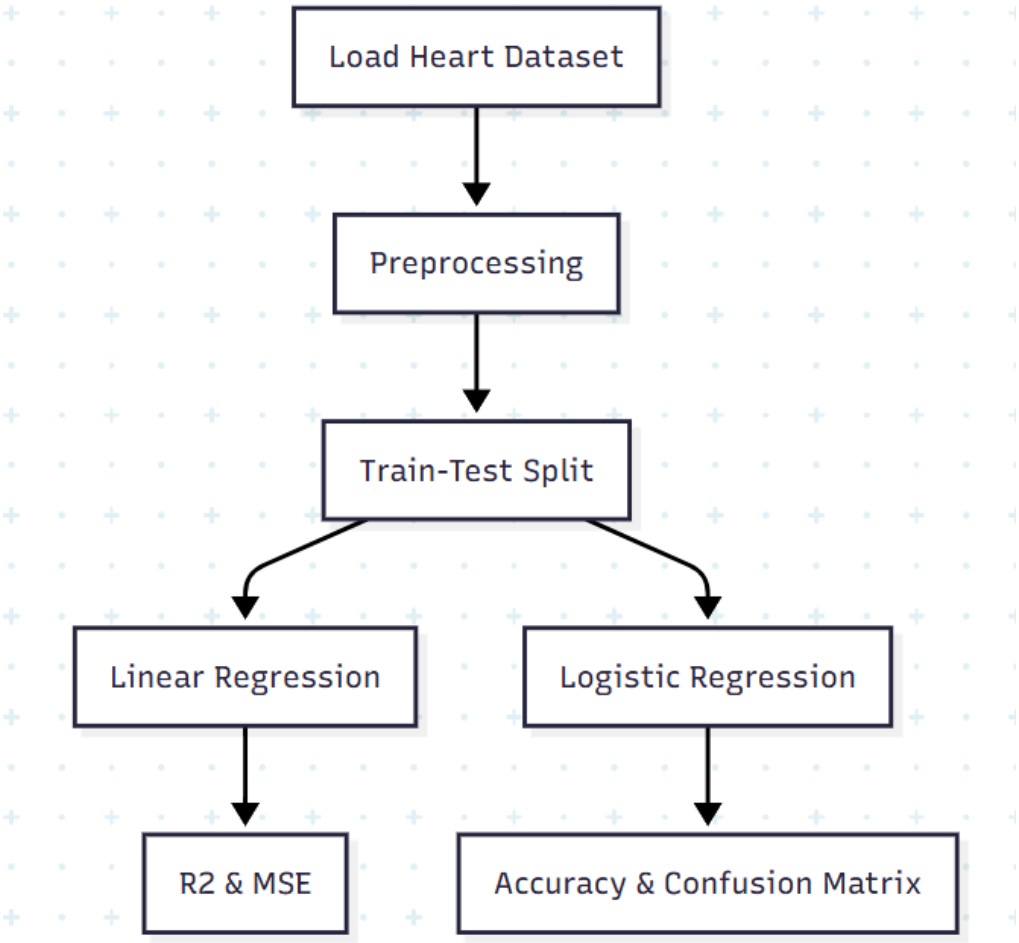
- Assumes linear decision boundary
- Requires feature scaling
- Performance decreases for highly complex relationships

---

5. Methodology / Workflow

1. Loaded Heart Disease dataset from Kaggle
2. Checked for missing values
3. Split dataset into training and testing sets
4. Applied Linear Regression to predict cholesterol (`chol`)
5. Evaluated using:
    - R^2 Score
    - Mean Squared Error
6. Applied Logistic Regression to predict heart disease (`target`)
7. Scaled features using StandardScaler
8. Evaluated using:
    - Accuracy
    - Confusion Matrix
    - Classification Report

🔁 Workflow Diagram



---

6. Performance Analysis

🔹 Linear Regression

- R^2 Score indicates how well the model explains variation in cholesterol levels.
- Mean Squared Error measures prediction error.

A moderate R^2 indicates the model captures some variation in cholesterol levels.

🔹 Logistic Regression

- Accuracy Score indicates classification correctness.
- Confusion Matrix shows true positives and true negatives.
- Precision and Recall evaluate classification reliability.

The model achieved high classification accuracy in predicting heart disease presence.

7.Hyperparameter Tuning

Linear Regression

- Default parameters used.
- No major hyperparameters.

Logistic Regression

- `max_iter` increased to ensure convergence.
- Features scaled using StandardScaler.
- Proper train-test split applied to avoid overfitting.

8.CODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.preprocessing import StandardScaler

# ----------------------------------------
# 1. Load Dataset
# ----------------------------------------
df = pd.read_csv("heart.csv")
print("Dataset Shape:", df.shape)

# Convert categorical columns
df = pd.get_dummies(df, drop_first=True)

# ========================================
# PART A: LINEAR REGRESSION
# Predict Cholesterol
# ========================================
```

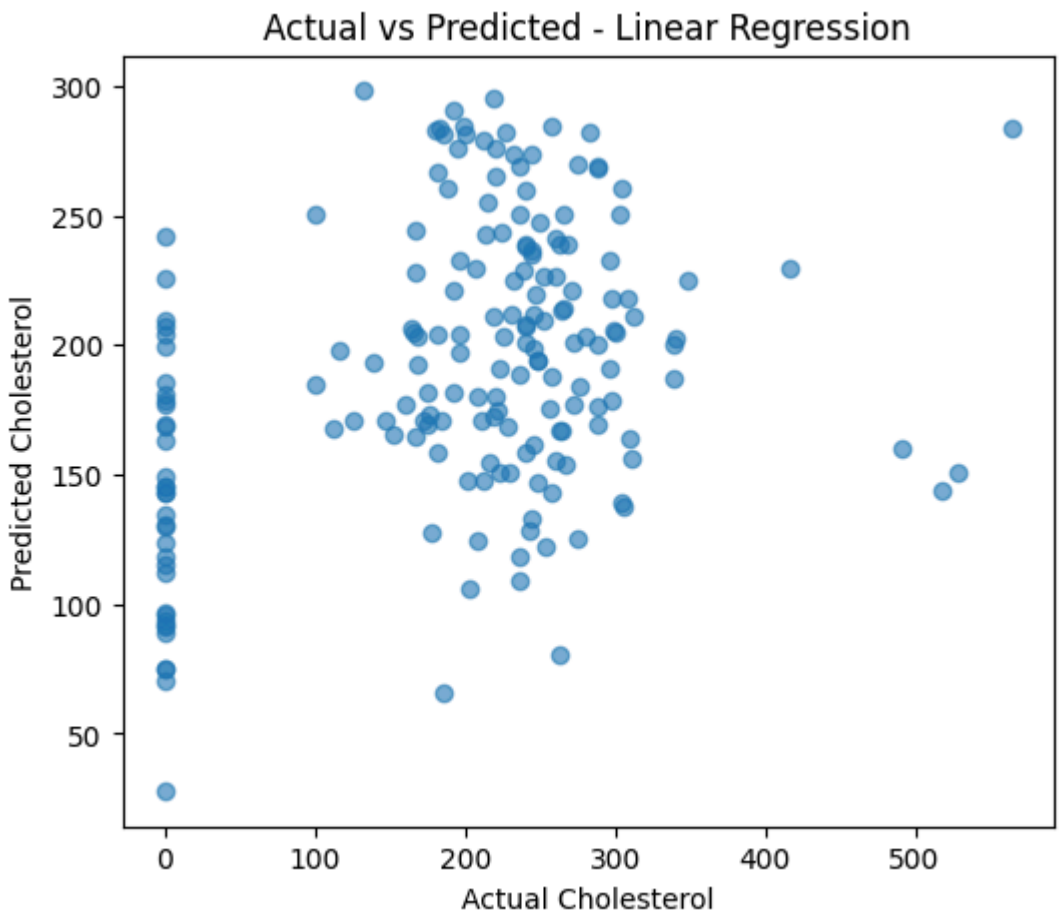
```python
X_lin = df.drop("Cholesterol", axis=1)
y_lin = df["Cholesterol"]

X_train, X_test, y_train, y_test = train_test_split(
    X_lin, y_lin, test_size=0.2, random_state=42
)

lin_model = LinearRegression()
lin_model.fit(X_train, y_train)

y_pred_lin = lin_model.predict(X_test)

r2 = r2_score(y_test, y_pred_lin)
mse = mean_squared_error(y_test, y_pred_lin)

print("\n--- Linear Regression Evaluation ---")
print(f"R2 Score: {r2:.4f}")
print(f"Mean Squared Error: {mse:.2f}")

# Optional Regression Plot
plt.figure(figsize=(6,5))
plt.scatter(y_test, y_pred_lin, alpha=0.6)
plt.xlabel("Actual Cholesterol")
plt.ylabel("Predicted Cholesterol")
plt.title("Actual vs Predicted - Linear Regression")
plt.show()

# ==========================================
# PART B: LOGISTIC REGRESSION
# Predict HeartDisease
# ==========================================

X_log = df.drop("HeartDisease", axis=1)
y_log = df["HeartDisease"]

scaler = StandardScaler()
X_log = scaler.fit_transform(X_log)

X_train, X_test, y_train, y_test = train_test_split(
    X_log, y_log, test_size=0.2, random_state=42
)
```

```
log_model = LogisticRegression(max_iter=5000)
log_model.fit(X_train, y_train)

y_pred_log = log_model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred_log)
cm = confusion_matrix(y_test, y_pred_log)

print("\n--- Model Evaluation ---")
print(f"Accuracy Score: {accuracy*100:.2f}%")
print("\nConfusion Matrix:\n", cm)

print("\nClassification Report:\n")
print(classification_report(y_test, y_pred_log))

# Confusion Matrix Heatmap
plt.figure(figsize=(6,5))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title("Confusion Matrix")
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.show()

print("\n--- Final Tuned Model Evaluation ---")
print(f"Accuracy Score: {accuracy*100:.2f}%")
print(classification_report(y_test, y_pred_log))
```

9.OUTPUT

Dataset Shape: (918, 12)

--- Linear Regression Evaluation ---
R2 Score: 0.1158
Mean Squared Error: 11459.87

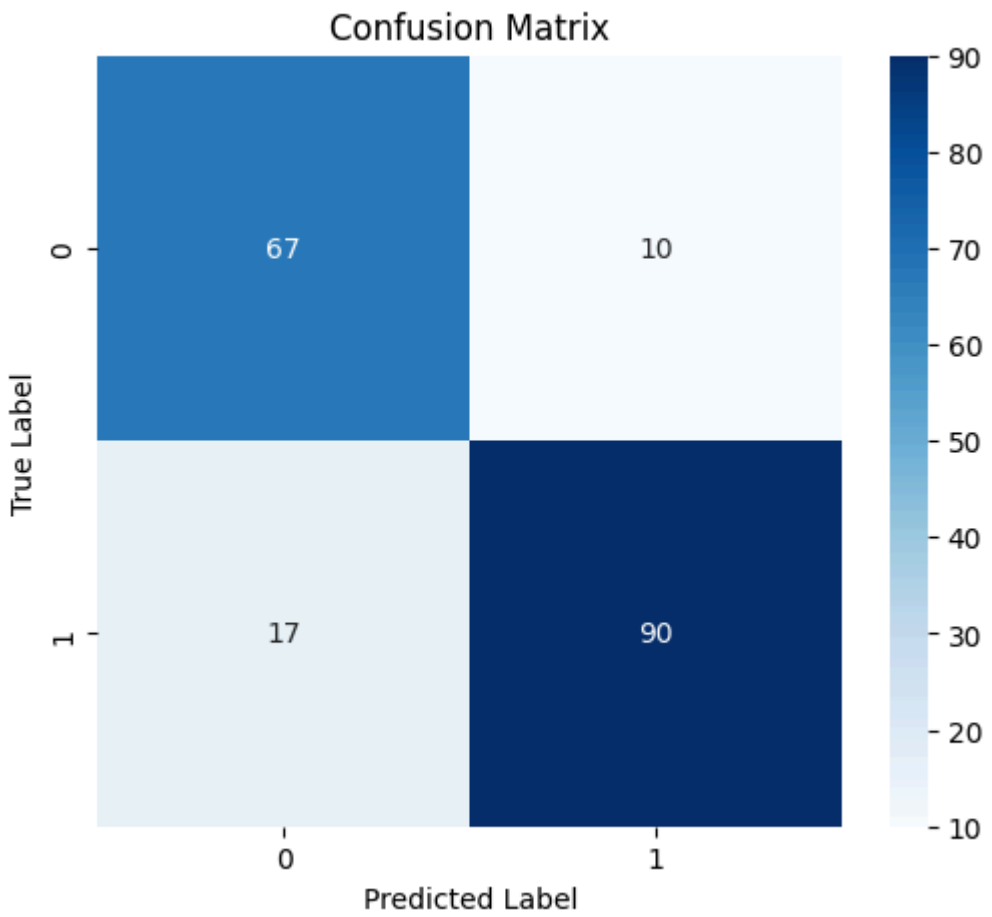


--- Model Evaluation ---
Accuracy Score: 85.33%

Confusion Matrix:
[[67 10]
[17 90]]

Classification Report:

	precision	recall	f1-score	support
0	0.80	0.87	0.83	77
1	0.90	0.84	0.87	107
accuracy			0.85	184
macro avg	0.85	0.86	0.85	184
weighted avg	0.86	0.85	0.85	184



--- Final Tuned Model Evaluation ---
Accuracy Score: 85.33%

	precision	recall	f1-score	support
0	0.80	0.87	0.83	77
1	0.90	0.84	0.87	107
accuracy			0.85	184
macro avg	0.85	0.86	0.85	184
weighted avg	0.86	0.85	0.85	184

10.Conclusion

Linear Regression was successfully used to predict cholesterol levels from medical attributes.

Logistic Regression effectively classified heart disease presence.

Both regression techniques were implemented using the same Kaggle Heart Disease dataset.

The experiment demonstrates practical application of regression models on real-world medical data.